stetrics & Gynecology confirmed that active smoking affects female fertility. Obesity, too, is a factor, with the risk of infertility being thrice as high in obese females, as a 2018 study published in *Reproductive Biology and Endocrinology* noted. Environmental toxins were found to be another villain, as another study in the same issue of the journal discovered. Particulate matter found in highly polluted cities was said to lead to reduced fertility. "It is natural for the number of eggs and quantity to reduce with age in women, but this is being aggravated by modern urban lifestyle choices," says Dr Chandrika Anand, Consultant, Obstetrics & Gynecology at the Fortis Hospitals in Nagarbhavi, Bengaluru.

However, despite these concerns, the market for infertility treatment and ART remains quite small in India, not least because of the costs involved as well as the fact that social taboos continue to cloud such discussions in India. The Indian infertility diagnosis and treatment market is expected to grow at a compound annual rate of just 12.5 per cent between 2022 and 2028, according to a study by Orion Market Research. Egg-freezing, therefore, is also a predominantly urban phenomenon. In fact, there is a parallel rise in the number of women who are having children before the age of 30. The International Institute for Population Sciences (IIPS), which conducts the National Family Health Survey (NFHS) in India, recently conducted a study that showed that against 35 per cent women in 1992-93 who had completed childbearing by 30, as many as 64 per cent women in the same age bracket had gone through childbirth by 2019-2021.



"If someone goes for egg-freezing, they must be aware that they have only IVF as an option, and usually at the same clinic where you freeze the eggs"

DR PREETI RASTOGI
Director, Institute of Obstetrics and Gynaecology, Medanta, Gurugram

A Welcome Start, But With Caveats

The growing conversation around egg-freezing is definitely a welcome start. But doctors advise caution and want those considering the option to keep a few crucial points in mind.

One, knowing what increases the risk of infertility is important to understand egg-freezing. The female egg is a specialised and sensitive cell. It is also more important than the sperm because it has the ability to correct sperm-related abnormalities during embryo development. "This is not to dismiss the importance of sperm quality, but to understand that pregnancy outcomes are better when good quality eggs are produced/fertilised as opposed to poor ones leading to poor embryo quality as well," says Dr Selvaraj.

The selection process of eggs during freezing or IVF is not under medical control, and hence, in order to get healthy ones that will lead to an eventual pregnancy, experts say women must be conscious of overall wellness prior to the process. "We do counselling sessions before egg-freezing and inform women of the factors that affect egg quality. Just because you freeze your eggs does not mean they will be healthy and fertile. So, women must be active, cut out junk food, quit smoking and alcohol in the months leading up to the egg-freezing," says Kolkata-based gynaecologist Dr Sharmila Lahiri.

Two, age is crucial. According to ISAR, the success rates of egg-freezing in India are higher for women in their 20s and early 30s than those who go for the procedure later in life. "The older the age of the egg, the higher the risk of chromosomal abnormalities, and, if undetected, they may result in either mental abnormalities or physical malformations, or sometimes both, which require termination as early as possible," says Dr Selvaraj.

Some doctors also advise women to freeze fertilised eggs if they have a partner. "If one is unmarried or without a partner, they can opt for egg-freezing. But if they are married, then embryo-freezing is recommended. The difference is that with the latter, you will have a healthy embryo with you. With egg-freezing, the concern remains whether the other partner's sperm will be healthy or whether it will fertilise or not. Just because you freeze an egg does not guarantee success in IVF," says Dr Surveen Ghumman Sindhu, senior director and head of department, Infertility & IVF, Max Hospital, Patparganj, New Delhi. There is also a legal caveat: while only women have legal rights over eggs, both partners have rights over frozen fertilised eggs.

Three, even though egg-freezing is generally safe, there are side-effects to consider, especially with the hormonal medication given before the procedure. Says Dr Lahiri, "I would say, try natural birth at the right age first. I don't think everyone should feel compelled to freeze eggs—it's expensive and we still don't know if there are absolutely no side-effects in the long term from the hormones given, mostly because a few years ago it wasn't even such a common process." Experts say the procedure should be seen as an option, not a solution. "It is an insurance to bank on (with limited assurances) if natural methods or assisted reproduction fails," says Dr Selvaraj. "One must discuss with doctors. We have to talk about how many years they want to delay pregnancy by, egg quality at the current age, chances of natural conception, make sure fertility won't be compromised at the time of conception. Also, if someone goes for egg-freezing, they need to know they have only IVF as an option and usually at the same clinic where you freeze the eggs," says Dr Rastogi.

However, regardless of whatever a person's eventual decision may be, the fact that a growing number of people are now willing to at least discuss options like egg-freezing and reproductive autonomy is a feat in itself. In that sense, this party has just started. ■